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Artificial Intelligence-Driven Mallakhamb Pose Detection and Automated Virtual Coaching System Using YOLOv8, MediaPipe, and Random Forest

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ABSTRACT Mallakhamb is a traditional Indian sport combining aerial yoga, gymnastics, and martial arts on a vertical wooden pole. Automated pose estimation in Mallakhamb presents severe computer vision challenges due to extreme body inversions, complex geometric postures, and frequent occlusion by the pole. This paper presents an end-to-end artificial intelligence framework for automated Mallakhamb pose detection, posture accuracy scoring, and real-time virtual coaching. The system incorporates a multi-stage pipeline comprising YOLOv8 object localization with dynamic 40% spatial margin auto-zooming, MediaPipe 3D pose landmark extraction (33 spatial points), and torso-centric translation and scale normalization. Biomechanical feature engineering is applied by computing 3D spatial joint angles for eight critical anatomical joints. Evaluating across a balanced dataset of 4,750 augmented images spanning 19 distinct Mallakhamb poses using 5-fold stratified cross-validation, the proposed Random Forest classifier achieves a state-of-the-art F1-score of 98.43%, outperforming XGBoost (97.93%), K-Nearest Neighbors (96.63%), and Support Vector Machines (95.61%). Furthermore, an automated scoring engine converts joint angular deviation against ideal biomechanical standards into a percentage accuracy score, accompanied by natural language and audio feedback. Deployed via a Streamlit web application, the framework offers an accessible, real-time digital coach for athletes and judges.

INDEX TERMS Biomechanical feedback, Computer vision, Machine learning, Mallakhamb pose estimation, MediaPipe, Random Forest, Sports analytics, Streamlit, YOLOv8.

I. INTRODUCTION

MALLAKHAMB is an ancient Indian sport where an athlete performs dynamic acrobatic postures, aerial yoga, and wrestling grips on a vertical polished wooden pole or hanging rope. Originating as a conditioning exercise for wrestlers in the 12th century, Mallakhamb has evolved into an internationally recognized competitive sport demanding extraordinary strength, agility, spatial awareness, and extreme flexibility [12]. In recent years, the integration of artificial intelligence (AI) and computer vision into sports analytics has revolutionized athlete training, injury prevention, and objective judging [3], [9], [10]. While computer vision models have achieved remarkable success in main-

stream sports such as football, basketball, and track events, applying pose estimation to traditional acrobatic disciplines like Mallakhamb remains fundamentally challenging [1]–[3]. Key difficulties include:

- 1) **Extreme Inversions and Unconventional Body Orientations:** Performers routinely hang upside down, twist along non-canonical axes, or lock themselves into intricate geometric shapes.
- 2) **Pole Occlusion and Background Interference:** The physical pole frequently obstructs major body segments (e.g., torso, thighs, shoulders), confusing standard human pose estimation networks.
- 3) **Scale and Distance Variations:** Unconstrained cam-

era positions lead to major scale variations as performers move vertically along the pole.

To overcome these challenges, this paper introduces an intelligent, end-to-end Mallakhamb Pose Detection and Virtual Coaching System. The system combines deep learning-based object detection, geometric skeleton extraction, scale-invariant feature normalization, ensemble machine learning classification, and real-time biomechanical feedback generation.

A. MAIN CONTRIBUTIONS

The primary contributions of this work are summarized as follows:

- **Robust Two-Stage Vision Pipeline:** We deploy a YOLOv8 object detector to crop performers with an adaptive 40% spatial margin, isolating the athlete from noisy backgrounds before passing the region of interest to MediaPipe 3D Pose [3], [5].
- **Scale and Translation Invariant Normalization:** We formulate a torso-centric coordinate transformation that translates body landmarks relative to the hip midpoint and scales coordinates by the Euclidean torso length, rendering the pipeline invariant to camera distance and athlete height [3].
- **Comprehensive Classifier Benchmarking:** We present a rigorous comparative evaluation of four machine learning models (Random Forest, XGBoost, Support Vector Machine, and K-Nearest Neighbors) evaluated via 5-fold stratified cross-validation over 19 distinct Mallakhamb pose classes [6]–[8], [11].
- **Biomechanical Scoring and Feedback System:** We design a quantitative joint angle deviation model comparing 8 core 3D joint angles against ideal posture vectors, generating real-time text and speech feedback through a Streamlit web interface [3], [9].

The rest of this paper is organized as follows: Section II reviews related literature. Section III describes the proposed methodology and architectural pipeline. Section IV details experimental results and model benchmarking. Section V discusses limitations, and Section VI concludes the paper.

II. RELATED WORK

Human pose estimation (HPE) has advanced rapidly with deep learning architectures. Landmark frameworks such as OpenPose [1], AlphaPose [2], and MediaPipe Pose [3] enable multi-person or single-person 2D and 3D skeleton tracking in real time. MediaPipe Pose, in particular, utilizes a lightweight two-stage detector-tracker framework providing 33 3D landmarks with high spatial resolution and computational efficiency suitable for edge deployment. In sports domain applications, pose estimation has been utilized for golf swing analysis, gymnastic vault scoring, yoga pose correction [9], and swimming technique assessment [10]. However, these systems rely primarily on upright human poses and standard background settings. In Mallakhamb and

aerial acrobatics, literature remains sparse. Existing studies either rely on manually annotated images or fail under extreme inversions due to model overfitting on conventional upright datasets. Our work bridges this gap by coupling a YOLOv8 auto-zoom detector with normalized 3D landmark geometric calculations and ensemble classifiers, providing an accurate, robust, and computationally lightweight virtual coach specifically tailored for Mallakhamb [3], [5]–[7].

III. METHODOLOGY AND SYSTEM ARCHITECTURE

The overall pipeline consists of six primary modules: Dataset Preprocessing & Augmentation, YOLOv8 Auto-Zoom Bounding Box Extraction, MediaPipe 3D Landmark Extraction, Invariant Coordinate Normalization, Biomechanical Joint Angle Computation, and the Classification & Virtual Coach Feedback Engine [3], [5], [6].

A. DATASET PREPROCESSING AND AUGMENTATION

The raw dataset initially contained 1,248 uncured images distributed unequally across 21 pose categories. To prepare a standardized benchmark:

- 1) **Class Merging:** Semantic duplication among minor variations was audited, consolidating the dataset into 19 distinct, structurally unique Mallakhamb poses.
- 2) **Data Augmentation:** To eliminate class imbalance and prevent overfitting, data augmentation techniques—including horizontal flipping, slight rotations ($\pm 15^\circ$), and brightness adjustments ($\pm 20\%$)—were applied. Each class was balanced to exactly 250 images, producing an augmented dataset of $N = 4,750$ images.

B. YOLOV8 OBJECT LOCALIZATION AND AUTO-ZOOM

To mitigate background noise and scale variations, we integrate YOLOv8 (nano variant, `yolov8n.pt`) [5]. Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, YOLOv8 detects the person bounding box $\mathbf{B} = [x_1, y_1, x_2, y_2]$. If multiple detections occur, the box with the maximum area $(x_2 - x_1) \times (y_2 - y_1)$ is selected as the primary performer. A spatial margin of 40% is added along the x - and y -axes to ensure limbs extending beyond the torso remain fully enclosed:

$$\Delta x = 0.40 \times (x_2 - x_1), \quad \Delta y = 0.40 \times (y_2 - y_1) \quad (1)$$

$$x'_1 = \max(0, x_1 - \Delta x), \quad x'_2 = \min(W, x_2 + \Delta x) \quad (2)$$

$$y'_1 = \max(0, y_1 - \Delta y), \quad y'_2 = \min(H, y_2 + \Delta y) \quad (3)$$

The cropped image $\mathbf{I}_{\text{crop}} = \mathbf{I}[y'_1 : y'_2, x'_1 : x'_2]$ is passed to the landmark extractor.

C. MEDIAPIPE LANDMARK EXTRACTION AND NORMALIZATION

Using MediaPipe Pose (`model_complexity=2`) [3], 33 body keypoints are extracted from $\mathbf{I}_{\text{crop}}$. Each keypoint $i \in \{0, \dots, 32\}$ comprises spatial coordinates $\mathbf{p}_i = [x_i, y_i, z_i, v_i]^T$, where x_i, y_i are normalized pixel positions,

z_i is landmark depth relative to the hip center, and v_i represents visibility confidence. Landmark extraction yielded 4,642 valid skeleton vectors across the dataset. To ensure invariant representation across varying camera distances and body proportions, coordinates undergo torso-centric translation and scaling:

- 1) **Hip Centering (Translation):** The hip midpoint $\mathbf{m}_{\text{hip}}$ is computed:

$$\mathbf{m}_{\text{hip}} = \frac{\mathbf{p}_{\text{left_hip}} + \mathbf{p}_{\text{right_hip}}}{2} \quad (4)$$

Every point is centered: $\mathbf{p}'_i = \mathbf{p}_i - \mathbf{m}_{\text{hip}}$ for spatial coordinates (x, y, z) .

- 2) **Torso Scaling:** The Euclidean length of the torso S_{torso} is calculated using the shoulder midpoint $\mathbf{m}_{\text{shoulder}}$:

$$\mathbf{m}_{\text{shoulder}} = \frac{\mathbf{p}_{\text{left_shoulder}} + \mathbf{p}_{\text{right_shoulder}}}{2} \quad (5)$$

$$S_{\text{torso}} = \|\mathbf{m}_{\text{shoulder}} - \mathbf{m}_{\text{hip}}\|_2 \quad (6)$$

- 3) **Normalized Feature Vector:** Each coordinate is scaled: $\mathbf{p}''_i = \mathbf{p}'_i / S_{\text{torso}}$. The flattened vector $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{132}$ serves as the input feature vector.

D. BIOMECHANICAL JOINT ANGLE COMPUTATION

Beyond raw 3D landmarks, 8 key 3D joint angles are calculated to assess posture correctness. For three anatomical points A, B, C where B is the joint vertex:

$$\vec{u} = \mathbf{p}_A - \mathbf{p}_B, \quad \vec{v} = \mathbf{p}_C - \mathbf{p}_B \quad (7)$$

$$\theta_{\text{joint}} = \arccos \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|_2 \|\vec{v}\|_2 + \epsilon} \right) \times \frac{180}{\pi} \quad (8)$$

where $\epsilon = 10^{-6}$ avoids division by zero. Angles are calculated for Left/Right Elbow, Left/Right Shoulder, Left/Right Hip, and Left/Right Knee.

E. CLASSIFICATION MODELS AND VIRTUAL COACHING ENGINE

Four machine learning algorithms were trained on $\mathbf{X}_{\text{norm}}$ to classify the 19 Mallakhamb poses: Random Forest (100 trees), XGBoost, Support Vector Machine (RBF kernel, $C = 10$), and K-Nearest Neighbors ($k = 5$) [6]–[8], [11]. Once a pose class $\hat{y}$ is predicted, the user's joint angles θ_{user} are compared against the ideal posture vector $\theta_{\text{ideal}}^{(\hat{y})}$. The Mean Absolute Error (MAE) across $K = 8$ joints is evaluated:

$$\text{MAE} = \frac{1}{K} \sum_{k=1}^K \left| \theta_{\text{user},k} - \theta_{\text{ideal},k}^{(\hat{y})} \right| \quad (9)$$

Accuracy score S_{acc} is mapped linearly, where an error of 30° corresponds to 0% score:

$$S_{\text{acc}} = \min \left(100.0, \max \left(0.0, 100.0 - \frac{\text{MAE}}{30.0} \times 100.0 \right) \right) \quad (10)$$

IV. EXPERIMENTAL RESULTS AND DISCUSSION

Experiments were conducted using Python 3.10, scikit-learn (v1.3.0), mediapipe (v0.10.9), and ultralytics (v8.1.0) [3], [5], [11]. Performance evaluation was carried out using 5-Fold Stratified Cross-Validation over 4,642 dataset samples.

A. CLASSIFIER PERFORMANCE COMPARISON

Table 1 summarizes the comparative performance metrics across all candidate models. Random Forest achieved the

TABLE 1. Comparative Performance of Classification Models (5-Fold Cross-Validation)

Model	Precision (%)	Recall (%)	F1-Score (%)	Latency (ms)
Random Forest	98.51	98.36	98.43	4.2
XGBoost	98.02	97.85	97.93	12.8
KNN ($k = 5$)	96.80	96.48	96.63	3.1
SVM (RBF Kernel)	95.88	95.35	95.61	8.5

highest overall performance with an F1-Score of **98.43%**, demonstrating superior generalization across all 19 Mallakhamb pose classes [6].

B. ABLATION STUDY: IMPACT OF YOLOV8 AUTO-ZOOM

An ablation experiment was conducted comparing raw full-frame MediaPipe landmark extraction versus YOLOv8-cropped MediaPipe extraction under varying subject distances (1.5 m to 5.0 m). As shown in Table 2, cropping with

TABLE 2. Landmark Extraction Success & Accuracy With / Without YOLOv8 Auto-Zoom

Distance (m)	Raw MediaPipe (%)	YOLOv8 + MediaPipe (%)	F1-Gain (%)
1.5	98.2	99.1	+0.9
3.0	84.5	98.6	+14.1
5.0	61.2	96.4	+35.2

a 40% margin dramatically improves landmark detection stability at long distances, boosting F1-score performance by over 35% at 5.0 m.

C. REAL-TIME VIRTUAL COACH WEB DEPLOYMENT

The finalized Random Forest model (`best_pose_classifier.py`) was integrated into a Streamlit web application [6], [11]. The interface displays real-time skeleton visual overlays, an accuracy score SVG gauge, coach feedback cards, and text-to-speech audio guidance.

V. LIMITATIONS

When an athlete's body is almost entirely wrapped around the rear side of the wooden pole, MediaPipe landmark confidence drops below threshold ($v_i < 0.5$). Additionally, the current system evaluates static frame snapshots rather than dynamic continuous fluidity during transition routines.

VI. CONCLUSION

This paper presented an intelligent end-to-end framework for Mallakhamb pose recognition, accuracy scoring, and virtual coaching. Combining YOLOv8 auto-zooming, MediaPipe

3D pose extraction, torso-based coordinate normalization, and Random Forest classification, the system achieved a benchmark F1-Score of **98.43%** across 19 distinct Mallakhamb poses [3], [5], [6]. Future work will focus on dynamic temporal sequence modeling using Spatial-Temporal Graph Convolutional Networks (ST-GCN).

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